

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(D_i \left[\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right] \right) \qquad - \nabla \cdot (\varepsilon \nabla \varphi) = \sum_i z_i F c_i$$

$$R_r = k_{0,r} \left[c_O e^{-\alpha_r F \eta_r / RT} - c_R e^{(1-\alpha_r) F \eta_r / RT} \right], \quad \eta_r = \varphi_m - \varphi_s - E_{,r}^\circ$$

$$\mu_i^{\text{steric}} = -\ln \theta, \quad \theta = 1 - \sum_i a_i c_i$$

$$\varepsilon \partial_n \varphi = C_S (V_{\text{app}} - \varphi_{\text{OHP}}), \quad C_S = 0.20 \text{ F/m}^2$$

$$c_i \approx c_i^{\text{bulk}} \exp(-z_i F \varphi / RT)$$

$$\lambda_D = \sqrt{\frac{\varepsilon RT}{2F^2I}} \ll L_{\text{eff}}, \qquad I = \frac{1}{2} \sum_i z_i^2 c_i$$

$$\mu_H = u_H + \frac{z_H F}{RT} \varphi, \quad c_H = \exp\left(\mu_H - \frac{z_H F}{RT} \varphi\right), \quad \mathbf{J}_H = -D_H c_H \nabla \mu_H$$

$$\log r_r^{(\text{cat})} = \ln k_{0,r} + u_O - \alpha_r \frac{F \eta_r}{RT}, \quad \log r_r^{(\text{anod})} = \ln k_{0,r} + u_R + (1 - \alpha_r) \frac{F \eta_r}{RT}, \quad R_r = r_r^{(\text{cat})} - r_r^{(\text{anod})}$$

$$c_b(x) = c_b^{\text{bulk}} \frac{\left(1 - \sum_{j \in \text{dyn}} a_j c_j\right) e^{-z_b F \varphi / RT}}{\theta_b + a_b c_b^{\text{bulk}} e^{-z_b F \varphi / RT}},$$

$$\theta_b = 1 - \sum_j a_j c_j^{\text{bulk}} - a_b c_b^{\text{bulk}}$$